J1 90814-0912
J2 90814-0912
J5 90814-0912
J6 90814-0912
J7 90814-0912
J8 90814-0912
J3 90814-0912
J4 90814-0912
HT
R8 18K
R7 18K
R3 18K
R4 18K
R5 18K
R6 18K
R1 18K
R2 18K
Nixie_1_K[0..9]
Nixie_2_K[0..9]
Nixie_3_K[0..9]
Nixie_4_K[0..9]
Nixie_5_K[0..9]
Nixie_6_K[0..9]
Nixie_7_K[0..9]
Nixie_8_K[0..9]
R13 98K
R14 98K
R15 98K
R16 98K
R9 98K
R10 98K
R11 98K
R12 98K
GND
Nixie_8_K2
Nixie_8_K1
Nixie_8_K0
J9 Debug
+3.3V
Ur1 HV5622PG-G
Ur2 HV5622PG-G
Ur3 HV5622PG-G
HVOUT1 HVOUT2 HVOUT3 HVOUT4 HVOUT5 HVOUT6 HVOUT7 HVOUT8 HVOUT9 HVOUT10 HVOUT11 HVOUT12 HVOUT13 HVOUT14 HVOUT15 HVOUT16 HVOUT17 HVOUT18 HVOUT19 HVOUT20 HVOUT21 HVOUT22 HVOUT23 HVOUT24 HVOUT25 HVOUT26 HVOUT27 HVOUT28 HVOUT29 HVOUT30 HVOUT31 HVOUT32
DATAOUT NC NC NC *POL CLK VSS VDD *LE DATAIN *BL NC
+12V GND
C1 0.1u
C2 0.1u
C3 0.1u
LS_data_out
LS_clk_out
LS_LE_out
LS_BL_out
Level shifters
J12 Debug1
LS_data_in
LS_clk_in
LS_LE_in
R17 4.7k
R18 4.7k
R19 4.7k
R20 4.7k
R22 10k
R25 10k
R24 10k
R26 10k
R27 3.6k
Q2 PMV37ENEAR
Q3 PMV37ENEAR
Q4 PMV37ENEAR
Q5 PMV37ENEAR
LS_BL_out low by default so that all shift register outputs are off
HV DCDC
J10 HV_EN
PWR_FLAG
Y1 AIUR-08 coil
D1 D_Schottky
J11 HT
C4 100uF
C6 0.1u
U1 MAX1771xSA
SHDN V+ EXT CS REF FB AGND GND
Q1 IPD60R280P7
C8 4.7u
C9 0.1u
R23 0R100
R21 1M5
RV1 5K
R28 10k
C7 0.1uF
Boot strapping pins
S1 Reset
S2 Boot
R29 10k
R31 10k
R32 10k
C12 1u
C16 0.1u
ESP_EN
PU_GPIO2
PU_GPIO8
boot_dl
U3 ESP32-C3-MINI-1
C10 0.1u
C11 10u
EN/CHIP_PU
3V3
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO21/U0TXD
GPIO20/U0RXD
GPIO19/USB_D+
GPIO18/USB_D-
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
GPIO4/ADC1_CH4
GPIO5/ADC2_CH0
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
mode_sel
J15 UART
Power & mode toggle switch
J13 3.3V_EN
U2 TLV76133DCY
VI VO GND
C14 10u
SW1 Tog_ON_ON_(ON)
C13 10u
C15 0.1u
R30 10k
J14 12V_IN
J16
mode_sel

Nixie clock controller
Theo Lephay
12/2025

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Title:

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